APPLIED MULTIVARIATE METHODS
833:525 / 970:527

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Homework 10 (50 points)

Problem 1:

Article to read: Miller et al. (2017), Response to Name in Infants Developing Autism Spectrum: a Prospective Study.

Please limit your response to each item below to 3-5 sentences.

a) **Describe the patient population. Your response should cover, as best as possible, the who, where, when, and what of the population of interest. (10 points)**

The patient population studied in Miller et al. (2017) consists of 156 infants, categorized into two main groups based on their risk for developing Autism Spectrum Disorder (ASD). The groups are:

1. High-risk infants - These are younger siblings of children already diagnosed with ASD. There were 96 infants in this category, 20 of whom were later diagnosed with ASD themselves.
2. Low-risk infants - These are younger siblings of children with typical development, without any family history of ASD. This group included 60 infants.

These participants were followed from 6 months to 24 months of age, with assessments at intervals (6, 9, 12, 15, 18, and 24 months). The study took place at the MIND Institute at the University of California, Davis. By 36 months, the children were classified into one of three outcome groups based on whether they developed ASD, were high-risk without ASD, or were low-risk without ASD.

b) **What was the purpose of the analyses? Your response should include mention of (read: list) the outcome, main predictor(s), and population of interest (recall: X, Y, P from Lectures 01 and 02) (10 points)**

The purpose of the analyses conducted in the study was to examine longitudinal patterns of response to name from 6 to 24 months of age in infants at high and low risk for Autism Spectrum Disorder (ASD). The analyses aimed to determine how early and consistently infants who develop ASD fail to respond to their names compared to those who do not develop ASD. The outcome, main predictors, and the population of interest as per the study framework are:

Outcome (Y): The primary outcome was the infants' response to their names, which was assessed during visits at 6, 9, 12, 15, 18, and 24 months of age. The outcome was whether infants oriented to their names as a binary outcome of pass or fail.

Main Predictors (X):

- Age: The age of the infants during each test session was a continuous predictor, looking at changes over time.
- Risk Group: The risk status (high-risk sibling of a child with ASD vs. low-risk sibling of typically developing child) served as a categorical predictor.
- ASD Development Status: Determined at 36 months based on standardized diagnostic criteria.

Population of Interest (P): The population of interest includes the High-risk infants i.e. younger siblings of children already diagnosed with ASD, and Low-risk infants i.e. younger siblings of children with typical development.

c) **Define the outcome, i.e., state what concept is being measured, how it is being measured including the measurement level with units or categories, and how the outcome is repeated in nature. (10 points)**

The outcome in the study by Miller et al. (2017) is the infants' response to their name, which is a behavioral marker that might indicate the development of Autism Spectrum Disorder (ASD). This response is assessed through an "orients to name" task where the infant's ability to orient towards the examiner when their name is called is observed and recorded. The task is conducted at multiple sessions over a

period from 6 to 24 months of age.

Measurement Details:

Concept Being Measured: The ability of infants to orient to their names when called, which serves as a potential early indicator of ASD.
How It Is Measured: The measurement is done using a binary scale where:
0 = Orients to name with eye contact on both calls, at least one of which is on the first trial
1 = Orients with eye contact at least once
2 = Does not orient on any trial
A score of 0 or 1 is considered "passing," and a score of 2 is considered a "failure."
Measurement Level and Units: The data are categorical and binary (pass/fail based on whether the score is 0-1 or 2).
Repeated Nature of the Outcome: The "orients to name" task is administered repeatedly during visits at 6, 9, 12, 15, 18, and 24 months of age, which allows for longitudinal tracking of the infants' responsiveness over time.
This repeated measure across multiple time points helps in understanding the developmental trajectory of each infant with respect to this specific behavioral response, and its potential links to ASD. Generalized Estimating Equations (GEEs): The study employed Generalized Estimating Equations (GEE) to analyze the longitudinal data. This statistical method is particularly suited for dealing with correlated data resulting from repeated measurements on the same subjects. Outcome Variable: The binary outcome (fail/pass) of the "Orients to Name" task was modeled.

Link Function and Variance Function: The models used a logit link function and binomial variance function, appropriate for binary outcome data.

Handling of Time: The analysis treated the age at which each assessment occurred as a continuous predictor, allowing the estimation of changes in the probability of failing the "Orients to Name" task over time. The interaction terms between age and group enabled the exploration of whether developmental trajectories in name responsiveness differed between groups.

Adjustments/Controlling: By including sex and adjusting for potential confounders, the study aimed to account for additional variability in response to name that could influence the outcome aside from risk group and age. The use of empirical (robust) variance estimators in the GEE approach helps to increase the reliability of the standard error estimates, which is crucial when the data structure involves repeated measures. The models facilitated hypothesis testing regarding the effects of risk group and age on the likelihood of failing the response to name task. P-values of less than .05 were considered statistically significant, indicating robust findings regarding the developmental trajectories of infants at different risk levels for ASD.

d) **What variables are included as predictors in the model? Read the first paragraph of the data analytic plan to help. (10 points)**

In the study, the variables included as predictors in the model are:

1. Group: This categorical variable distinguishes between the different groups of infants involved in the study. The groups are:
- Low-risk group without ASD
- High-risk group without ASD
- Group with ASD outcomes

2. Visit Age: This continuous predictor variable represents the age of the infants in months at each assessment visit. The variable is centered at 6 months to examine changes over time from this baseline. 3. Child Sex: This categorical variable accounts for the biological sex of the infants.

e) **What did the investigators find? Interpret model estimate(s) of interest. (10 points)**

The investigators discovered significant variations over time in how infants responded to their names, based on their risk levels. The Generalized Estimating Equations (GEE) model highlighted key findings from 6 to 24 months of age:

Low-Risk without ASD vs High-Risk without ASD: Infants who later developed ASD consistently showed more failures in orienting to their names from 9 months of age compared to those in the low-risk group without ASD, with this gap widening as they grew older. A similar pattern was observed when comparing infants diagnosed with ASD against high-risk infants without ASD. The ASD group's likelihood of failing the task was consistently higher and became more pronounced over time.

Effect of Visit Age:

Developmental Trend: In both the low-risk and high-risk groups without ASD, there was a developmental trend where the likelihood of failing the task decreased with age, indicating typical developmental progress in responding to their names.

ASD Group Trend: In contrast, the ASD group's trajectory remained constant, showing no improvement with age, which diverges from the typical developmental trajectory observed in the other groups.

Model Estimates:

Interaction and Differences: The model revealed a significant interaction between visit age and group **($\chi 2(2) = 8.86$, $P = .01$)**, suggesting that the improvement rates in name responsiveness differed across groups. Notably, the ASD group did not exhibit the expected developmental gains in responding to names.

Odds Ratios: The odds ratios demonstrated that the probability of failing to respond to names was considerably higher in the ASD group compared to the low-risk group across all ages assessed, with an especially strong effect size at 24 months (OR approximately 18).

Interpretation of Model Estimates:

Early Indicator: The findings suggest that difficulties with the name response task as early as 9 months could indicate potential ASD development. These challenges persist over time, unlike the improvements observed in typically developing and high-risk but non-ASD children.

Developmental Implications: The consistent trajectory in the ASD group suggests a lack of natural developmental advances in social communication skills typical for this age, potentially highlighting fundamental neurodevelopmental differences associated with ASD.

These results emphasize the importance of vigilant early monitoring and possibly early intervention for infants who show persistent challenges with responding to their names, particularly those in high-risk groups. Early detection of these developmental delays allows for targeted early interventions aimed at improving social communication skills before more pronounced symptoms of ASD become established.